

CRISTIAN ASSAIANTE

Curriculum Vitae

PERSONAL DATA

NATIONALITY:	Italian	LOCATION:	Rome, Italy
PERSONAL EMAIL:	assaiente.cris@gmail.com	PHONE NUMBER	+39 3403244996
WORK EMAIL:	assaiente@diag.uniroma1.it	PERSONAL PAGE:	cristianassaiente.github.io

EMPLOYMENT

Nov. 2024 - Now	Research Fellowship <i>Sapienza University of Rome</i> Collaborating with Google on compiler testing
Nov. 2021 - Oct. 2024	Ph.D. Scholarship <i>Sapienza University of Rome</i>
Jan. 2020 - Jul. 2023	Organizer and Training Coordinator CYBERCHALLENGE.IT for <i>Sapienza University of Rome</i> unit
Spring 2021	Training Coordinator and Challenges Author OLICYBER.IT

EDUCATION

Nov. 2021 - Jan. 2026:	Ph.D. in ENGINEERING IN COMPUTER SCIENCE <i>Sapienza University of Rome</i> Thesis: Symbolic Debugging of Optimized Code: Measuring, Testing, Tuning and Enhancing Debug Information Quality Advisor: Prof. Leonardo Querzoni Co-Advisor: Dr. Daniele Cono D'Elia Reviewers: Prof. Zhendong Su and Prof. Cristiano Giuffrida
Oct. 2019 - Oct. 2021	M.Sc. in ENGINEERING IN COMPUTER SCIENCE <i>Sapienza University of Rome</i> Final grade: 110/110 with honors (<i>summa cum laude</i>) Thesis: A Study of the Completeness of Debug Symbols in Optimizing Compilers Advisor: Dr. Daniele Cono D'Elia
Oct. 2016 - Jul. 2019	B.Sc. in COMPUTER AND CONTROL ENGINEERING <i>Sapienza University of Rome</i> Final grade: 110/110 with honors (<i>summa cum laude</i>) Thesis: A Micro-Architectural Red Pill Advisor: Dr. Daniele Cono D'Elia
Sep. 2011 - Jul. 2016	High School Diploma in INFORMATION TECHNOLOGY <i>I.T.I.S. A. Pacinotti, Fondi</i> Final grade: 100/100

RESEARCH INTERESTS

My main research interests are compilers, program analysis, and software testing. I am particularly interested in developing practical techniques that improve debuggability, reliability and performance of modern software systems. My work spans symbolic debugging of optimized code, profile-guided optimizations and program analysis applied in diverse contexts, with a strong emphasis on methods that integrate seamlessly into real-world toolchains and software development workflows. I aim to bridge the gap between programming languages, software engineering and systems security by designing techniques that help developers build more robust and secure software without sacrificing performance. I am particularly drawn to challenging problems where existing solutions remain impractical or insufficient.

PUBLICATIONS

Conferences

- 2026 **C. Assaiante**, S. Di Biasio, S. Kumar, G. A. Di Luna, D. C. D'Elia and L. Querzoni. *Towards Threading the Needle of Debuggable Optimized Binaries*. In Proceedings of the 24th ACM/IEEE International Symposium on Code Generation and Optimization (CGO '26). [CORE26 RANK: A]
- 2024 **C. Assaiante**, S. Nicchi, D. C. D'Elia and L. Querzoni. *Evading Userland API Hooking, Again: Novel Attacks and a Principled Defense Method*. In Proceedings of the 21st Conference on Detection of Intrusions and Malware & Vulnerability Assessment (DIMVA '24). [CORE26 RANK: B]
- 2023 **C. Assaiante**, D. C. D'Elia, G. A. Di Luna and L. Querzoni. *Where Did My Variable Go? Poking Holes in Incomplete Debug Information*. In Proceedings of the 28th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS '23). [CORE26 RANK: A*]

TEACHING

- 2026 **Adjunct Professor** for *Sistemi di Calcolo* (6 ECTS module) course, Spring 2026. B.Sc. in Computer and Control Engineering at Sapienza University of Rome
- 2024 **Adjunct Professor** for *Sistemi di Calcolo* (3 ECTS module) course, Spring 2024. B.Sc. in Computer and Control Engineering at Sapienza University of Rome
- 2023 **Adjunct Professor** for *Sistemi di Calcolo* (3 ECTS module) course, Spring 2023. B.Sc. in Computer and Control Engineering at Sapienza University of Rome
- 2022 **Teaching Assistant** for *Sistemi di Calcolo* course, Spring 2022. B.Sc. in Computer and Control Engineering at Sapienza University of Rome

SERVICE

Conferences

- 2026 Shadow Program Committee member for **ICSE '27**
Artifact Evaluation Committee member for **ACM CCS '26** and **NDSS '27**
- 2025 Artifact Evaluation Committee member for **ACM CCS '25**, **NDSS '26** and **USENIX Security '25**
- 2024 Artifact Evaluation Committee member for **NDSS '25**
- 2022 Shadow Program Committee member for **EuroSys '23**
Artifact Evaluation Committee member for **EuroSys '23**

Journals

- 2023 Reviewer for **Computers & Security** (COSE)
Reviewer for **SoftwareX** (SOFTX)

HONORS AND AWARDS

- 2024 **Finalist** with *mhackeroni*, DEF CON 32 CTF Finals. Las Vegas, USA
- 2023 **Winner** with *mhackeroni*, HACK-A-SAT 4 CTF Finals. Las Vegas, USA
Finalist with *mhackeroni*, DEF CON 31 CTF Finals. Las Vegas, USA
- 2022 **Awarded** the BLACKHat ASIA *Student Scholarship*. Singapore
- 2021 **Selected** for the M.Sc. in Engineering in Computer Science *Honors Program* at Sapienza University of Rome
Awarded the 42ND IEEE SYMPOSIUM ON SECURITY AND PRIVACY *Student Travel Grant*. Online
Winner with *TheRomanXploit* (TRX), CSAW '21 Embedded Cybersecurity Challenge Finals. Online
Finalist with *mhackeroni*, DEF CON 29 CTF Finals. Las Vegas, USA
- 2020 **Awarded** the 15th CLUSIT thesis prize "*Innovare la sicurezza delle Informazioni*" ("Innovating Information Security") to *A Micro-Architectural Red Pill*. Online
Awarded the BLACKHat EU cybersecurity conference *Student Scholarship*. Online
Winner with *TheRomanXploit* (TRX), CSAW '20 Embedded Cybersecurity Challenge Finals. Online
Finalist with *mhackeroni*, DEF CON 28 CTF Finals. Las Vegas, USA
- 2019 **Selected** for the B.Sc. in Computer and Control Engineering *Honors Program* at Sapienza University of Rome
Winner with *TheRomanXploit* (TRX), CSAW '19 Embedded Cybersecurity Challenge Finals. Grenoble, France
Finalist with *mhackeroni*, DEF CON 27 CTF Finals (5th place). Las Vegas, USA

FUNDING

2022 **Ph.D. starting grant** from Sapienza University of Rome for the project: Practical Control-Flow Integrity for Software Security

TALKS AND PRESENTATIONS

2026 **C. Assaiante**, S. Di Biasio, S. Kumar, G. A. Di Luna, D. C. D'Elia and L. Querzoni. *Towards Threading the Needle of Debuggable Optimized Binaries*. Presented at **CGO '26** (Sydney, Australia) and as a tech talk for Google (Online)

2025 **C. Assaiante**. *API Monitoring in Malware Analysis*. Presented at the Malware Analysis and Incident Forensics (**MAIF**) course, M.Sc. in Cybersecurity at Sapienza University of Rome (Rome, Italy)

2024 **C. Assaiante**, S. Nicchi, D. C. D'Elia and L. Querzoni. *Evading Userland API Hooking, Again: Novel Attacks and a Principled Defense Method*. Presented at **DIMVA '24** (Lausanne, Switzerland), and at the Italian Conference on Cybersecurity **ITASEC '24** (Salerno, Italy)

2023 **C. Assaiante**, D. C. D'Elia, G. A. Di Luna and L. Querzoni. *Where Did My Variable Go? Poking Holes in Incomplete Debug Information*. Presented at **ASPLOS '23** (Vancouver, Canada)

SUMMER SCHOOLS

2023 6th Programming Language Implementation Summer School (**PLISS '23**). Bertinoro, Italy

2022 5th Advanced Online & Onsite Course on Data Science & Machine Learning (**ACDL '22**). Castelnuovo Berardenga, Italy

SKILLS

SOFT SKILLS: collaborative research, technical communication, teaching and mentoring

PROGRAMMING: Python, C, C++, x86/64 Assembly, Bash, Rust

TOOLS: LLVM, Linux, GDB, LLDB, IDA Pro, Ghidra, x64dbg, Docker, Git

LANGUAGES

ENGLISH: Fluent

ITALIAN: Native